



Advanced C Programming Lab

Session VII – Apr 30

Objectives

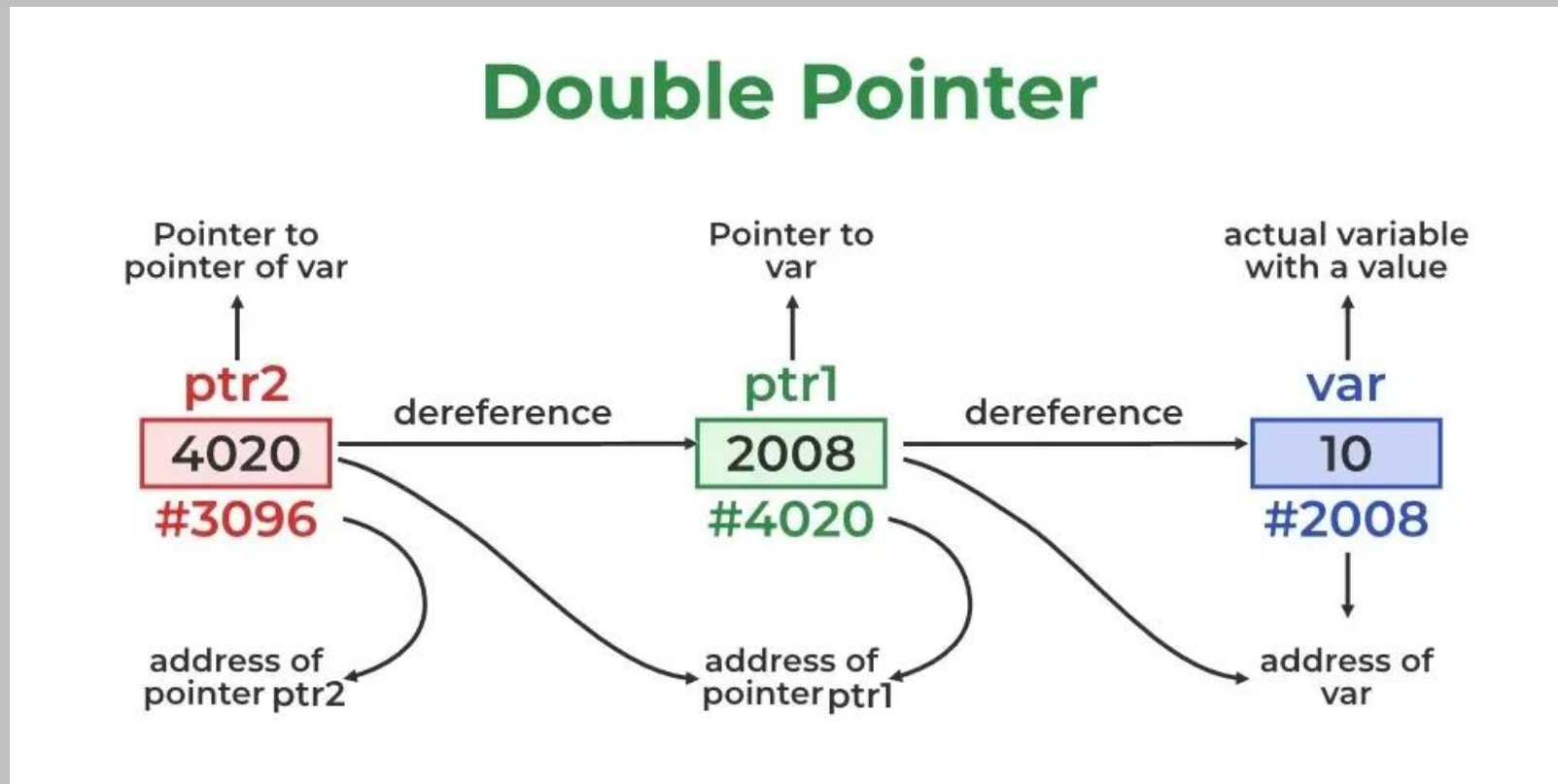
We will be talking about;

- Pointers to Pointers

In C, we can also have a pointer that points to another pointer which is called pointer to pointer (Double Pointer).

Pointer to pointer

Simply put, a normal pointer is like a note that has the address on it. A double pointer is another note that can tell you where the first note is kept.



Pointer to pointer

```
#include <stdio.h>
```

```
int main() {  
    int Num = 12;        // normal variable  
    int *ptr = &Num;     // pointer to int  
    int **pptr = &ptr;   // pointer to pointer  
  
    printf("Num = %d\n", Num);  
    printf("*ptr = %d\n", *ptr);  
    printf("**pptr = %d\n", **pptr);  
  
    return 0;  
}
```

Output:

```
Num = 12  
*ptr = 12  
**pptr = 12
```

Pointer to pointer

```
int Num = 12;  
int *ptr = &Num;  
int **pptr = &ptr;
```

- **Num** holds the value **12**.
- **ptr** holds the address of **Num**.
- **pptr** holds the address of **ptr**.
- ***ptr** gives the value of **Num**.
- ****pptr** also gives the value of **Num**, by going through both pointers.

Pointer to pointer

```
#include <stdio.h>
```

```
int main() {  
    int Num = 12;  
    int *ptr = &Num;  
    int **pptr = &ptr;  
  
    printf("Old Num = %d\n", Num);  
  
    **pptr = 34;    // changes Num  
  
    printf("New Num = %d\n", Num);  
    return 0;  
}
```

Output:

Old Num = 12
New Num = 34

Pointer to pointer

Passing Array of Strings to Function: Array of strings are generally stored as array of pointer to strings. This can be passed using double pointers to function..

```
#include <stdio.h>
// Function that takes array of strings as argument
void print(char **arr, int n) {
    for (int i = 0; i < n; i++)
        printf("%s\n", *(arr + i));
}
int main() {
    char *arr[10] = {"Hi", "Hello", "Salut", "Bonjour"};
    print(arr, 3);
    return 0;
}
```

Output:

Hi
Hello
Salut

Pointer to pointer

Size of a double pointer is same as size of any other pointer as it also stored address.

```
#include <stdio.h>
```

```
int main(){  
    // Defining single and double pointers  
    int a = 7;  
    int *ptr = &a;  
    int **d_ptr = &ptr;  
    // Size of double pointer  
    printf("%d bytes\n", sizeof(d_ptr));  
    // Size of a single pointer  
    printf("%d bytes", sizeof(ptr));  
    return 0;  
}
```

Output:

8 bytes
8 bytes

Thank you for you time and attention!

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Subject: C programming

References:

- <https://www.w3schools.com/>
- <https://www.geeksforgeeks.org/>
- <https://www.tutorialspoint.com/>